

CSE 3102 Lab 10 Task

1. Create a database for managing a departmental store named MS Saha & Brothers:

```
CREATE DATABASE MS_Saha_Brothers;
```

2. Enter these Tables:

```
CREATE TABLE Product (  
    productid INT PRIMARY KEY,  
    stock INT,  
    buying_price DECIMAL(10,2),  
    selling_price DECIMAL(10,2),  
    buying_date DATE,  
    selling_date DATE  
);
```

```
CREATE TABLE Customer (  
    customer_id INT PRIMARY KEY,  
    customer_name VARCHAR(100),  
    total_spent DECIMAL(10,2),  
    customer_due DECIMAL(10,2)  
);
```

```
CREATE TABLE Workers (  
    worker_id INT PRIMARY KEY,  
    worker_name VARCHAR(100),  
    salary DECIMAL(10,2),  
    worker_due DECIMAL(10,2)  
);
```

```
CREATE TABLE Rent (  
    security_money DECIMAL(10,2) PRIMARY KEY,  
    monthly_rent DECIMAL(10,2),  
    rent_due DECIMAL(10,2)  
);
```

```
CREATE TABLE growth (  
    record_id SERIAL PRIMARY KEY,  
    month DATE NOT NULL UNIQUE,  
    profit DECIMAL(10,2),  
    rent_due DECIMAL(10,2)  
);
```

```
CREATE TABLE Purchase (  
    purchase_id SERIAL PRIMARY KEY,  
    customer_id INT REFERENCES Customer(customer_id),  
    product_id INT REFERENCES Product(productid),  
    quantity INT,  
    amount DECIMAL(10,2)  
);
```

3. Insert data into the Database:

```
INSERT INTO Product VALUES  
(101, 50, 80.00, 120.00, '2023-01-15', NULL),  
(102, 30, 50.00, 90.00, '2023-02-10', NULL),  
(103, 100, 30.00, 60.00, '2023-03-05', NULL);
```

```
INSERT INTO customer (customer_id, customer_name, total_spent, customer_due)  
VALUES  
(1, 'Rahim Khan', 3500.00, 500.00),  
(2, 'Sima Akter', 2800.00, 0.00),  
(3, 'Faruq Hassan', 4200.00, 1200.00),  
(4, 'Tahmina Begum', 1500.00, 300.00),  
(5, 'Kamal Ahmed', 5300.00, 0.00);
```

```
INSERT INTO Workers VALUES  
(201, 'Abdul Malik', 20000.00, 0),  
(202, 'Nusrat Jahan', 18000.00, 0);
```

```
INSERT INTO Rent VALUES  
(5000.00, 15000.00, 0);
```

```
INSERT INTO growth (month, profit, rent_due)  
VALUES  
( '2023-06-01', 25000.00, 5000.00),  
( '2023-07-01', 31200.00, 4500.00),  
( '2023-08-01', 28750.00, 6000.00);
```

```
INSERT INTO Purchase (customer_id, product_id, quantity, amount) VALUES  
(1, 101, 2, 240.00),  
(2, 102, 1, 90.00),  
(3, 103, 5, 300.00);
```

Queries and Their Corresponding Answer:

1. Products with Profit Margin > 30%

Answer:

```
SELECT productid,  
       ROUND((selling_price - buying_price) / buying_price * 100, 2) AS profit_margin  
FROM Product  
WHERE (selling_price - buying_price) / buying_price > 0.3;
```

				productid	profit_margin
☐	 Edit	 Copy	 Delete	101	50.00
☐	 Edit	 Copy	 Delete	102	80.00
☐	 Edit	 Copy	 Delete	103	100.00

2. Total Inventory value:

Answer:

```
SELECT ROUND(SUM(stock * buying_price), 2) AS total_inventory_value  
FROM Product;
```

total_inventory_value

8500.00

3. Customer with pending dues:

Answer:

```
SELECT customer_name, customer_due  
FROM Customer  
WHERE customer_due > 0;
```

				customer_name	customer_due
☐	 Edit	 Copy	 Delete	Rahim Khan	500.00
☐	 Edit	 Copy	 Delete	Faruq Hassan	1200.00
☐	 Edit	 Copy	 Delete	Tahmina Begum	300.00

4. Average worker salary:

```
SELECT ROUND(AVG(salary), 2) AS avg_salary  
FROM Workers;
```

avg_salary

19000.00

5. Total Sales revenue:

Answer:

```
SELECT ROUND(SUM(amount), 2) AS total_revenue  
FROM Purchase;
```

total_revenue

630.00

6. Most Purchased Product (by quantity)

Answer:

```
SELECT p.productid, SUM(pu.quantity) AS total_quantity  
FROM Purchase pu  
JOIN Product p ON pu.product_id = p.productid  
GROUP BY p.productid  
ORDER BY total_quantity DESC  
LIMIT 1;
```

productid	total_quantity
-----------	----------------

103	5
-----	---

7. Products Never Sold:

Answer:

```
SELECT productid  
FROM Product  
WHERE productid NOT IN (SELECT DISTINCT product_id FROM Purchase)  
;; No products that are never sold
```

8. Top customer by spending and dues

Answer:

```
SELECT
```

```
    c.customer_id,  
    c.customer_name,  
    COALESCE(SUM(pu.amount), 0) AS total_spent,  
    c.customer_due
```

```
FROM Customer c
```

```
LEFT JOIN Purchase pu ON c.customer_id = pu.customer_id
```

```
GROUP BY c.customer_id, c.customer_name, c.customer_due
```

```
ORDER BY total_spent DESC
```

```
LIMIT 3;
```

customer_id	customer_name	total_spent ▼ 1	customer_due
3	Faruq Hassan	300.00	1200.00
1	Rahim Khan	240.00	500.00
2	Sima Akter	90.00	0.00

9. Customer purchase frequency:

```
SELECT
```

```
    c.customer_id,  
    c.customer_name,  
    COUNT(pu.purchase_id) AS purchase_count
```

```
FROM Customer c
```

```
LEFT JOIN Purchase pu ON c.customer_id = pu.customer_id
```

```
GROUP BY c.customer_id, c.customer_name;
```

customer_id	customer_name	purchase_count
1	Rahim Khan	1
2	Sima Akter	1
3	Faruq Hassan	1
4	Tahmina Begum	0
5	Kamal Ahmed	0

10. Products Above Stock (5 units):

Answer:

SELECT productid, stock

FROM Product

WHERE stock > 5;

← T →				productid	stock
☐	 Edit	 Copy	 Delete	101	50
☐	 Edit	 Copy	 Delete	102	30
☐	 Edit	 Copy	 Delete	103	100

